Stereo Vision: Disparity and Triangulation

The big idea: two eyes give you depth

Two cameras (or two eyes) a short distance apart see the same scene from slightly different spots, and that difference tells you how far away things are.

Try it now. Hold one finger at arm's length. Close your left eye, then swap to your right eye, back and forth. Your finger jumps sideways against the background. A tree far outside the window barely moves. The nearer the object, the bigger the jump.

That jump is the heart of stereo vision. Two words to learn:

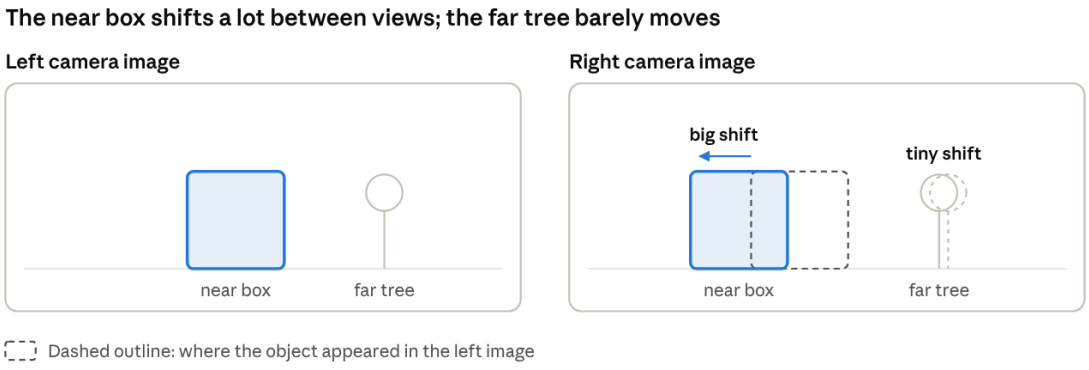
- **Disparity**: how far a point shifts sideways between the left image and the right image. For cameras, we count it in pixels.
- **Triangulation**: using the disparity and the camera geometry to calculate the real distance to the point.

The whole method in four steps:

1. Take one picture with the left camera and one with the right.
2. Find the same point (say, the corner of a box) in both pictures.
3. Measure how many pixels it shifted. That is the disparity.
4. Triangulate: turn the disparity into a distance.

Disparity: the sideways shift

Disparity is how far a point slides sideways between the left image and the right image.



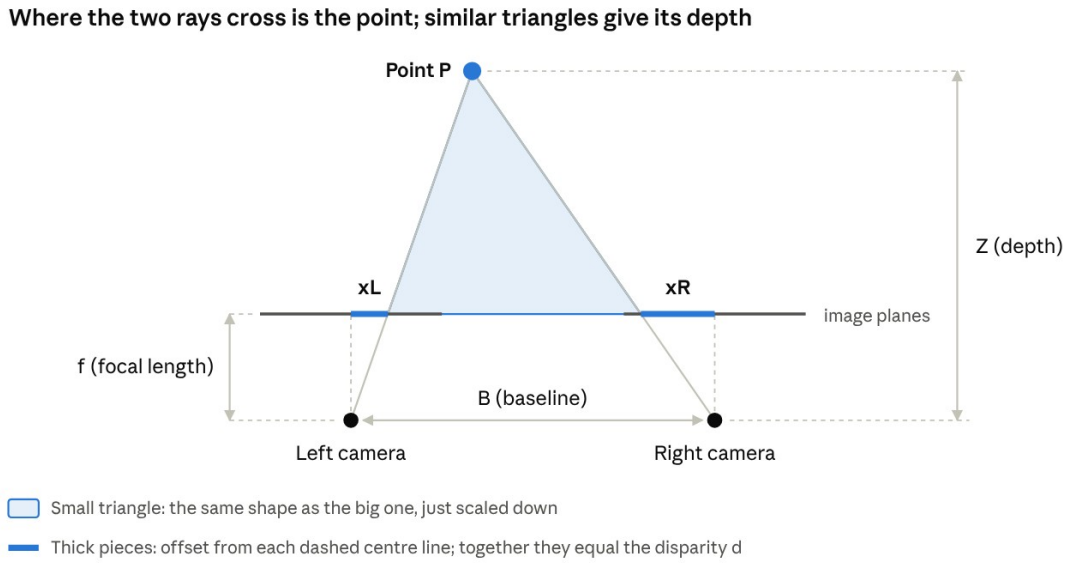
two views of one scene · near box vs far tree

In the right image the box has slid left by a large amount while the tree has hardly moved. Those two shifts are the two disparities.

- **Near objects** have large disparity. **Far objects** have small disparity.
- **A point at infinity** has zero disparity: it sits at the same pixel in both images.
- **A disparity map** is a grey image where each pixel's brightness is its disparity, so bright means near and dark means far.

Triangulation: from shift to distance

Triangulation means fixing a point's position from two known corners (the cameras) and the directions they see it in. Each camera gives one ray; the point sits where the rays cross.

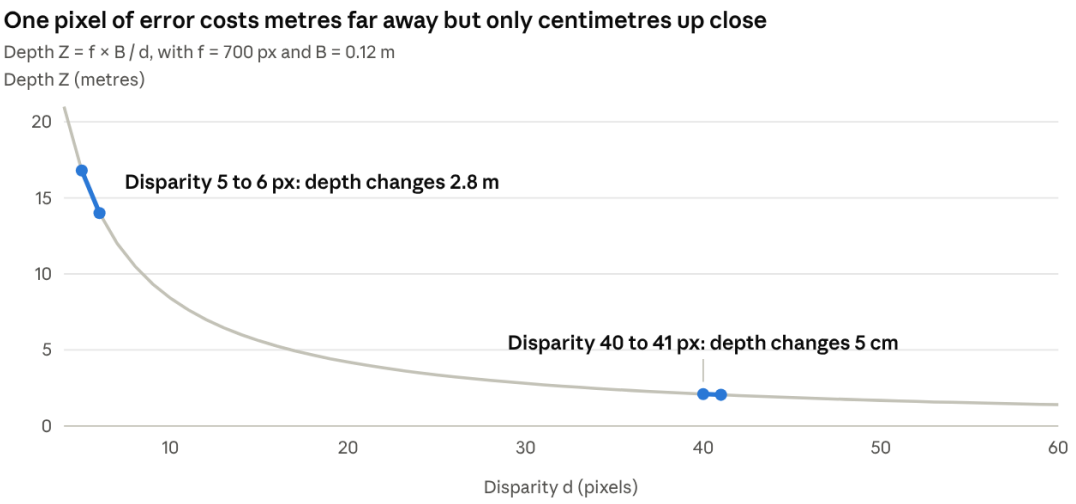


top-down view of two cameras and one point · not to scale

The picture looks down from above. The big triangle has the baseline B as its base and the depth Z as its height. The rays also cut a smaller triangle between the image planes and the point. It has the same apex and the same shape, so its sides are in the same ratios.

Baseline and distance: why far depth is shaky

Depth is least certain when disparity is small, because a one-pixel matching mistake then changes the answer by metres.



Rule of thumb: for a fixed one-pixel error, doubling the distance makes the depth error about four times bigger, and doubling the baseline cuts it in half. That is why the baseline matters.

	Short baseline	Long baseline
Accuracy far away	Worse: disparities are tiny	Better: disparities are larger
Close range	Good: both cameras see the object	Poor: the object may fall outside one view
Matching	Easier: the two views look alike	Harder: the views differ more
Suits	Close-range, hand-sized rigs	Long-range sensing

Finding the same point in both images

The formula is easy; the hard part is matching. For each pixel in the left image, the computer must find its twin in the right image. This is the **correspondence problem**.

The shortcut: rectification. When two cameras sit at the same height and point the same way, a point's twin lies on the *same row* of the other image. The search shrinks from the whole image to one row. Real cameras are never perfectly aligned, so software first warps both images to behave as if they were. Calibration (usually with a chessboard pattern) gives it the numbers to do that.

Where matching goes wrong:

Problem	Why it hurts	What you see
Blank wall or sky	Every patch looks the same	Holes or noise in the depth
Repeating pattern (fence, tiles)	Many equally good matches	Wrong depth, often in stripes
Shiny or glass surface	Looks different to each camera	Mismatches

Problem	Why it hurts	What you see
Occlusion	A point is visible to only one camera	No match exists
Poor calibration	Rows are not aligned	Noisy or failed search

Recap and cheat sheet

Stereo vision turns a pixel shift between two images into a distance, using one formula.

The pipeline:

1. Calibrate the cameras (find f, B and the lens distortion).
2. Rectify both images so matching points share a row.
3. Match points to get a disparity for each pixel.
4. Convert disparity to depth with $Z = f \times B / d$.

Remember:

- Disparity is a shift in pixels. Near objects shift a lot; far objects shift a little.
- Depth is inversely proportional to disparity. Double the shift, halve the distance.
- A wider baseline gives better accuracy at long range, but near objects may fall outside the overlap of the two views.
- Disparity of zero means the point is effectively at infinity.
- Keep units straight: f and d in pixels, B and Z in metres.